ECO 362: Financial Investments
Discounting

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Riskless assets

- Examples of riskless assets.
  - Banks: Savings deposits and certificates of deposit.
  - US Treasury: Bills (1-year maturity or less), notes (1–10 year maturity), and bonds (longer than 10-year maturity).
  - Some municipal bonds.
- What does riskless mean?
  - Nothing is truly riskless.
  - More precisely, safest assets available (i.e., negligible default risk).
  - Promises to pay back in nominal value (i.e., subject to inflation risk).

- Which of the following are riskless?
  - Savings account at Bank of America.
  - US Treasury bonds (rated AA+ by S&P).
  - Microsoft bonds (rated AAA by S&P).
- Different riskless assets could trade at different prices because of
  - Maturity.
  - Liquidity (i.e., how easy it is to resell in the secondary market).

Interest rate

- Interest rate is usually positive.
- Investor preferences imply that a \$1 today is worth more than a \$1 in the future.
- If the interest rate is negative, you would rather hold cash (at zero interest).
- In extraordinary circumstances, interest rates can become negative.
    - Quantitative easing: The central bank buys large quantities of bonds, driving up their prices.
    - Risk-averse institutions may perceive government bonds as safer than other options (e.g., deposits in other institutions).

Compounding frequency

- How frequently does interest accrue per year?
- Suppose that the annual interest rate is 5%.
- Semi-annual: Interest of $5\%/2 = 2.5\%$ every 6 months.
    - US Treasury.
- Monthly: Interest of $5\%/12 = 0.4167\%$ per month.
    - Savings accounts and mortgages.
- Daily: Interest of $5\%/365 = 0.0134\%$ per day.
    - Credit cards and student loans.

Compounding

- Suppose the annual interest rate is Y.
- Starting with \$1, how much do you have in T years?
- With annual compounding,

$$(1+Y)^T$$

- With semi-annual compounding,

$$(1+Y/2)^{2T}$$

- More generally, interest could be compounded n times per year.

$$(1+Y/n)^{nT}$$

- As $n \to \infty$, we have continuous compounding.

$$\exp(yT)$$

Conversion between annual and continuous compounding

- Y is the annually compounded interest rate.
- y is the continuously compounded interest rate.
- The relation between the two is

$$1 + Y = \exp(y)$$

- Continuous to annual:

$$Y = \exp(y) - 1$$

- Annual to continuous:

$$y = \ln(1 + Y)$$

Example 1: Compounding

- Illustrate the importance of compounding frequency by asking how long does it take to double your money (i.e., from \$1 to \$2)?
- Assume an annual interest rate of 5%.
- Annually compounded,

$$(1+0.05)^T = 2 \Rightarrow T = \frac{\ln(2)}{\ln(1+0.05)} = 14.21$$

- Semi-annually compounded,

$$(1+0.05/2)^{2T} = 2 \Rightarrow T = \frac{\ln(2)}{2\ln(1+0.05/2)} = 14.04$$

- Continuously compounded,

$$\exp(0.05T) = 2 \Rightarrow T = \frac{\ln(2)}{0.05} = 13.86$$

- It takes less time with more frequent compounding.

Discounting

- Compounding tells us how \$1 today grows after T years.

$$(1 + Y/n)^{nT}$$

- We can ask the opposite question. How much do you need to invest today to end up with \$1 after T years?

$$P(1 + Y/n)^{nT} = 1$$

Or

$$P = \frac{1}{(1 + Y/n)^{nT}}$$

- With continuous compounding,

$$P = \exp(-yT)$$

Example 1 (continued)

- With an annual interest rate of 5% over 10 years, how much do you need to invest today to end up with \$1?
- Annually compounded,

$$P = \frac{1}{(1+0.05)^{10}} = 0.6139$$

- Semi-annually compounded,

$$P = \frac{1}{(1+0.05/2)^{2\times 10}} = 0.6103$$

- Continuously compounded,

$$P = \exp(-0.05 \times 10) = 0.6065$$

- Need to invest less today with more frequent compounding.

Zero-coupon bond

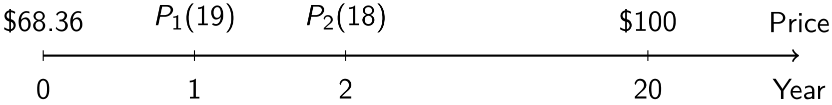
- A T-year zero-coupon bond (a.k.a. pure discount bond) pays \$1 in year T.
- Face value (or principal) is \$1.
- The price is related to the yield through the definition:

$$P_b(T) = \frac{1}{(1 + Y_b(T))^T}$$

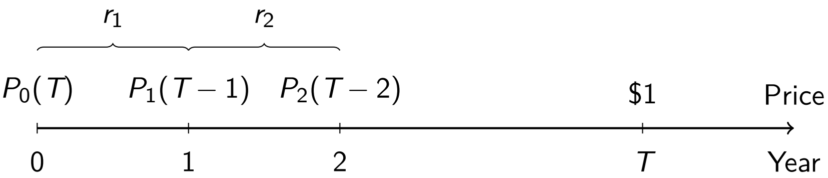
  - T: Maturity.
  - $P_b(T)$: Price.
  - $Y_b(T)$: Yield to maturity or annual interest rate.
- Unless stated otherwise, we will follow the convention that interest is annually compounded.

Is a zero-coupon bond risky?

- Consider a 20-year zero-coupon bond with \$100 face value.
- Riskless if your investment horizon is 20 years.
  - Pay price $P_0(20) = \$68.36$ today.
  - Receive \$100 in year 20.
- However, the price in between (e.g., in year 10) is uncertain.
- So risky if your investment horizon is less than 20 years.



Return



- Define the continuously compounded return from year 0 to 1 as

$$\exp(r_1) = \frac{P_1(T-1)}{P_0(T)}$$

- More generally, the return from $t-1$ to t is

$$\exp(r_t) = \frac{P_t(T-t)}{P_{t-1}(T-t+1)}$$

Example 2: 2-year zero-coupon bond

\$0.90	\$0.96	\$1	Price
0	1	2	Year

- The continuously compounded yield today is

$$y = \frac{1}{2} ln\left(\frac{1}{0.90}\right) = 5.27\%$$

- The returns in years 1 and 2 are

$$r_1 = ln\left(\frac{0.96}{0.90}\right) = 6.45\%$$

$$r_2 = ln\left(\frac{1}{0.96}\right) = 4.08\%$$

- The average return is

$$\frac{1}{2}(r_1 + r_2) = 5.27\%$$

An interpretation of yield

- Yield is a conventional way to quote the price.
- However, the yield on a zero-coupon bond has an economic interpretation as the average return if held to maturity.
- Rewrite the definition of yield as

$$\begin{aligned}\exp(y(T)) &= \left(\frac{1}{P_0(T)}\right)^{1/T} \\ &= \left(\frac{P_1(T-1)}{P_0(T)}\frac{P_2(T-2)}{P_1(T-1)}\cdots\frac{1}{P_{T-1}(1)}\right)^{1/T} \\ &= \left(\prod_{t=1}^{T}\exp(r_t)\right)^{1/T}\end{aligned}$$

- Taking the logarithms of both sides,

$$y(T) = \frac{1}{T}\sum_{t=1}^{T} r_t$$

Definition of riskless depends on investment horizon

- If you buy a 20-year zero-coupon bond, the total return over 20 years is certain.
- However, the annual returns before maturity (in years 1, 2,...) are random.
- Thus, a 20-year zero-coupon bond delivers a riskless return if your investment horizon is 20 years, but not if your investment horizon is shorter.
- For example,
  - To save to buy a house in 5 years, the riskless asset is a 5-year zero-coupon bond.
  - To save for retirement in 30 years, the riskless asset is a 30-year zero-coupon bond.

Short selling a bike

- There is a bike shortage on campus at the beginning of semester.
- Ben wants to take advantage of a temporary price spike.
- Ben borrows Janet's bike with a promise to return an equivalent bike (same model and condition) in 6 weeks.
- Today: Ben sells Janet's bike.
- In 6 weeks: Ben buys an equivalent bike and returns it to Janet.

Short selling a bond

- Janet owns a zero-coupon bond.
- Ben borrows Janet's bond with a promise to return the bond at maturity.
- Ben sells the borrowed bond immediately at market price P.
- Ben's cash flows are P today and $-\$1$ in year T.
- Economically, shorting a bond is equivalent to a loan. Ben borrows P today and promises to repay \$1 in the future.

Arbitrage

- Suppose two bonds with identical cash flows (i.e., same coupons and principal) trade at different prices.
- There is an arbitrage opportunity.
- Buy the cheaper bond and short the more expensive bond, pocketing the difference as an immediate profit.
- Since the cash flows are identical, the promised repayment on the short position is covered exactly by the cash flows from the long position.
- That is, you pocket an immediate profit today and have zero cash flows at all future dates.

Theory of no arbitrage

- The theory of no arbitrage says that two assets (more generally, portfolios) with identical cash flows must have the same price.
- The logic is that speculators will buy up the cheap asset (driving up its price) and short the expensive asset (driving down its price) until the two assets have the same price.
- Arbitrage happens so quickly that they are rarely observed in practice.
  - Do you ever see a $100 bill on a sidewalk?
- Benchmark theory that is not perfect but generates powerful predictions and has empirical support.

Annuity

- A T-year annuity with face value \$1 pays \$1 in each year from 1 to T.
- The price is related to the yield through the definition:

$$P_a(T) = \sum_{t=1}^{T} \frac{1}{(1 + Y_a(T))^t}$$

- The cash flows of an annuity are the same as a portfolio of zero-coupon bonds: 1 unit of each maturity from 1 to T years.

Price an annuity by no arbitrage

- No arbitrage implies that an annuity should have the same price as a portfolio of zero-coupon bonds with identical cash flows.
- Using our notation for price $P_b(t)$ and yield $Y_b(t)$ of t-year zero-coupon bonds,

$$
\begin{aligned}
P_a(T) &= \sum_{t=1}^{T} P_b(t) \\
&= \sum_{t=1}^{T} \frac{1}{(1 + Y_b(t))^t}
\end{aligned}
$$

- This equation is an example of the present-value formula. The price of a riskless asset is equal to its cash flows discounted by the zero-coupon yield curve.

Do not confuse yield vs. present value!

- The yield $Y_a(T)$ on an annuity is a way to quote its price.

$$P_a(T) = \sum_{t=1}^{T} \frac{1}{(1 + Y_a(T))^t}$$

  - This is a definition that is always true.
  - The yield is sometimes called the internal rate of return (in a corporate finance class).

- For example, a 2-year annuity with a yield of 0.65%:

$$1.98 = \frac{1}{1.0065} + \frac{1}{(1.0065)^2}$$

- The present-value formula prices an annuity relative to the zero-coupon yield curve.

$$P_a(T) = \sum_{t=1}^{T} \frac{1}{(1+Y_b(t))^t}$$

  - This formula is based on the theory of no arbitrage (which may not hold exactly in practice).
- For example, the zero-coupon yield is 0.42% at 1 year and 0.76% at 2 years. Pricing a 2-year annuity:

$$1.98 = \frac{1}{1.0042} + \frac{1}{(1.0076)^2}$$

Coupon bond

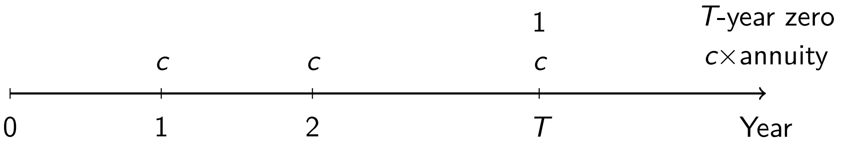
- A T-year coupon bond with coupon rate c and face value \$1 pays \$$c$ in each year from 1 to T and \$1 in year T.
- The price is related to the yield through the definition:

$$P_c(T) = \sum_{t=1}^{T} \frac{c}{(1 + Y_c(T))^t} + \frac{1}{(1 + Y_c(T))^T}$$

Two ways to price a coupon bond

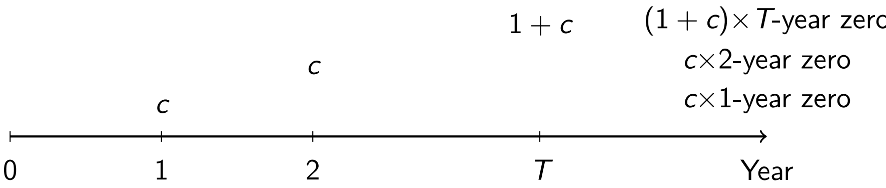
1. A T-year coupon bond is c units of a T-year annuity plus a T-year zero-coupon bond.

$$\begin{aligned} P_c(T) =& cP_a(T) + P_b(T) \\ =& \sum_{t=1}^{T} \frac{c}{(1+Y_b(t))^t} + \frac{1}{(1+Y_b(T))^T} \end{aligned}$$



2. A T-year coupon bond is a portfolio of zero-coupon bonds: c units of each maturity from 1 to $T-1$ years and $1+c$ units of maturity T.

$$
\begin{aligned}
P_c(T) &= \sum_{t=1}^{T-1} cP_b(t) + (1+c)P_b(T) \\
&= \sum_{t=1}^{T-1} \frac{c}{(1+Y_b(t))^t} + \frac{1+c}{(1+Y_b(T))^T}
\end{aligned}
$$



Present-value formula

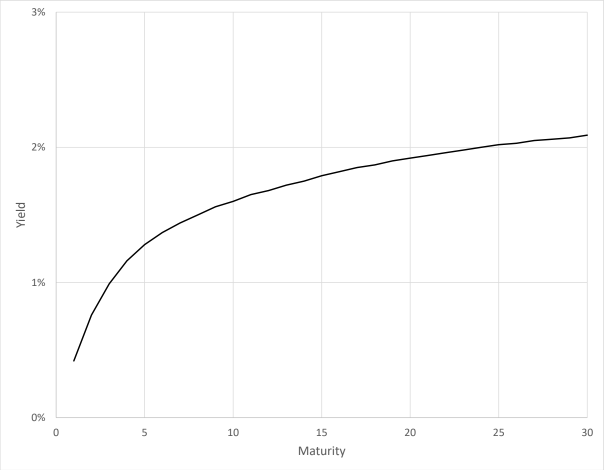
- The set of yields on zero-coupon bonds are called the zero-coupon yield curve or the term structure of riskless interest rates.
- Generalizing the examples of annuities and coupon bonds, you can use the term structure to discount any stream of riskless cash flows.
- For an asset that has riskless cash flow X_t in each year t,

$$P = \sum_{t=1}^{T} \frac{X_t}{(1 + Y_b(t))^t}$$

Treasury STRIPS

- Treasury bills are zero-coupon bonds, but notes and bonds are coupon bonds.
- STRIPS (Separate Trading of Registered Interest and Principal of Securities): Coupons are separated from the principal.
- For a newly issued 10-year note, there are coupon payments every 6 months and a principal repayment in year 10.
- Thus, STRIPS generates 20 zero-coupon bonds from the coupons (maturities 0.5, 1, 1.5,..., 10 years) and the principal (10-year zero-coupon bond).

US Treasury zero-coupon yield curve on 12/31/2021



Source: www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html

Example 3

- Start with the zero-coupon yield curve on 12/31/2021.
- Use Excel to price annuities and coupon bonds between maturities 1 to 30 years.

$$
\begin{aligned}
P_a(T) &= \sum_{t=1}^{T} \frac{1}{(1+Y_b(t))^t} \\
P_c(T) &= cP_a(T) + P_b(T)
\end{aligned}
$$

- Use the IRR (internal rate of return) function to compute yields.

US Treasury constant maturity yields (12/31/2021)

Maturity	Yield (%)
1 month	0.06
3 month	0.06
6 month	0.19
1 year	0.39
2 year	0.73
3 year	0.97
5 year	1.26
7 year	1.44
10 year	1.52
20 year	1.94
30 year	1.90

Source: www.federalreserve.gov/releases/h15/

Par bond

- A T-year par bond is a special case of the coupon bond when the coupon rate $c(T)$ equals the yield.
- The price of a par bond is equal to the face value.
- When the face value is \$1, the yield $c(T)$ on a par bond satisfies

$$1 = \sum_{t=1}^{T} \frac{c(T)}{(1+c(T))^t} + \frac{1}{(1+c(T))^T}$$

Par yield curve

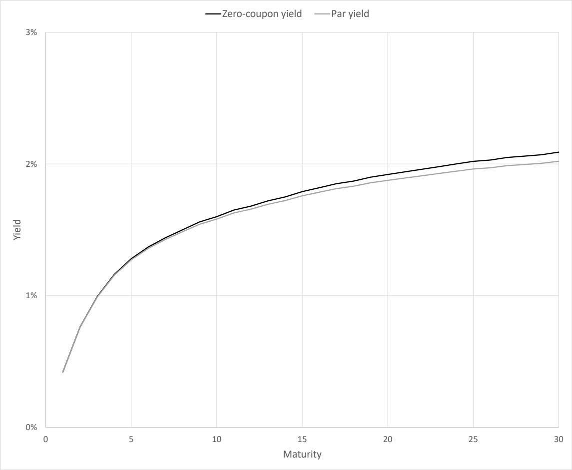
- When the Treasury issues a note/bond, they set the coupon rate so that it sells approximately at face value. Not exact because
    - Coupon rates are set in increments of 0.125%.
    - Treasury cannot predict market prices perfectly.
- Par yield at maturity T is the coupon rate $c(T)$ at which a T-year coupon bond would trade at face value.

$$
\begin{aligned}
1 &= \sum_{t=1}^{T} \frac{c(T)}{(1+Y_b(t))^t} + \frac{1}{(1+Y_b(T))^T} \\
&= \sum_{t=1}^{T} c(T) P_b(t) + P_b(T)
\end{aligned}
$$

- Solving for the par yield,

$$
c(T) = \frac{1 - P_b(T)}{\sum_{t=1}^{T} P_b(t)}
$$

Example 4: Par yield curve on 12/30/2021

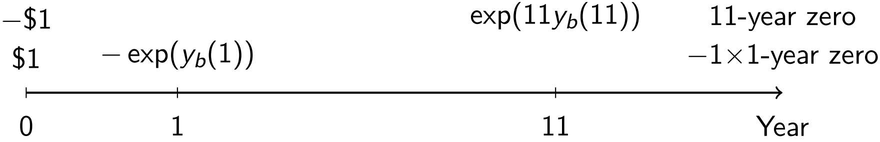


Forward contracts

- A forward contract locks in an interest rate (called a forward rate) on a future loan.
- Examples:
  1. Agree to buy a house in December 2021 with closing in March 2022. Lock in a mortgage rate from March 2022 to March 2052.
  2. Admitted to grad school in December 2021. Lock in a student loan rate from September 2022 to September 2032.

Example 5: Forward rate

- The zero-coupon yield curve on 12/31/2021.
    - 1-year zero: $y_b(1) = \ln(1.0042)$.
    - 11-year zero: $y_b(11) = \ln(1.0164)$.
- What is the forward rate from 12/2022 to 12/2032?
- Replicate the forward contract through a portfolio of zero-coupon bonds.
    1. Short \$1 of 1-year zero-coupon bond.
    2. Long \$1 of 11-year zero-coupon bond.



- The forward rate is the interest rate on this strategy.

$$\exp(10 y_f(1,11)) = \frac{\exp(11 y_b(11))}{\exp(y_b(1))}$$

- Taking the logarithm, the continuously compounded forward rate is

$$\begin{aligned} y_f(1,11) =& \frac{1}{10}(11 y_b(11) - y_b(1)) \\ =& \frac{1}{10}(11 \ln(1.0164) - \ln(1.0042)) = 1.76\% \end{aligned}$$

Summary

- Present-value formula: Discount any stream of riskless cash flows by the zero-coupon yield curve.
- Definition of riskless depends on the investment horizon.

Next steps

Key limitations of the theory that we will address in future classes.

1. Many assets have risky cash flows. How do you compute the price of a risky asset?
   - Generalize the theory of no arbitrage to risky cash flows.
2. The theory predicts how riskless assets are priced relative to the zero-coupon yield curve. But what determines the zero-coupon yield curve?
   - Term structure model: Expectation of future interest rates.